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(54) **MULTI-ANTENNA MOBILE PRINTERS**

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B41J 29/393 (2006.01)
H01Q 1/12 (2006.01)
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H01Q 23/00 (2006.01)
H05K 5/00 (2006.01)

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CPC . **B41J 3/36** (2013.01); **B41J 3/38** (2013.01);
B41J 29/02 (2013.01); **B41J 29/393** (2013.01); **H01Q 1/125** (2013.01); **H01Q 1/2258** (2013.01); **H01Q 1/2291** (2013.01);
H01Q 23/00 (2013.01); **H05K 5/0018** (2022.08); **B41J 2029/3932** (2013.01)

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See application file for complete search history.

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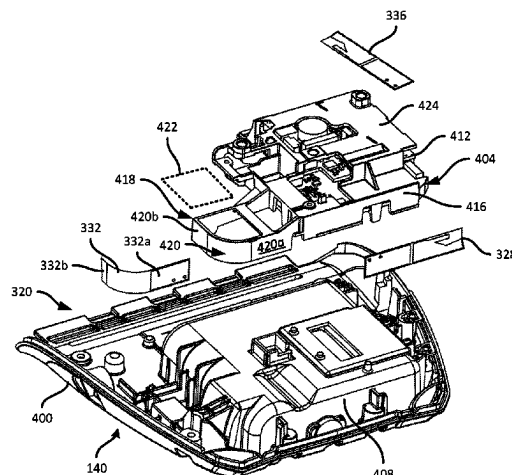
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(57) **ABSTRACT**

A printer includes: a housing with walls enclosing a compartment, and defining an adjacent media supply chamber; a cover coupled to an end of the housing beside the media supply chamber, movable between open and closed positions to expose or enclose the media supply chamber; a print head to apply indicia to media from the media supply chamber, and cooperate with the cover to form an outlet at a front wall of the housing for dispensing the media; a frame within the compartment, defining a first mounting surface beside and oriented towards a second end of the housing, a second mounting surface with a portion orthogonal to the first mounting surface and oriented towards a side of the housing, and a third mounting surface oriented towards the front wall; and first, second, and third antennas disposed on the first, second, and third mounting surfaces respectively.

15 Claims, 10 Drawing Sheets



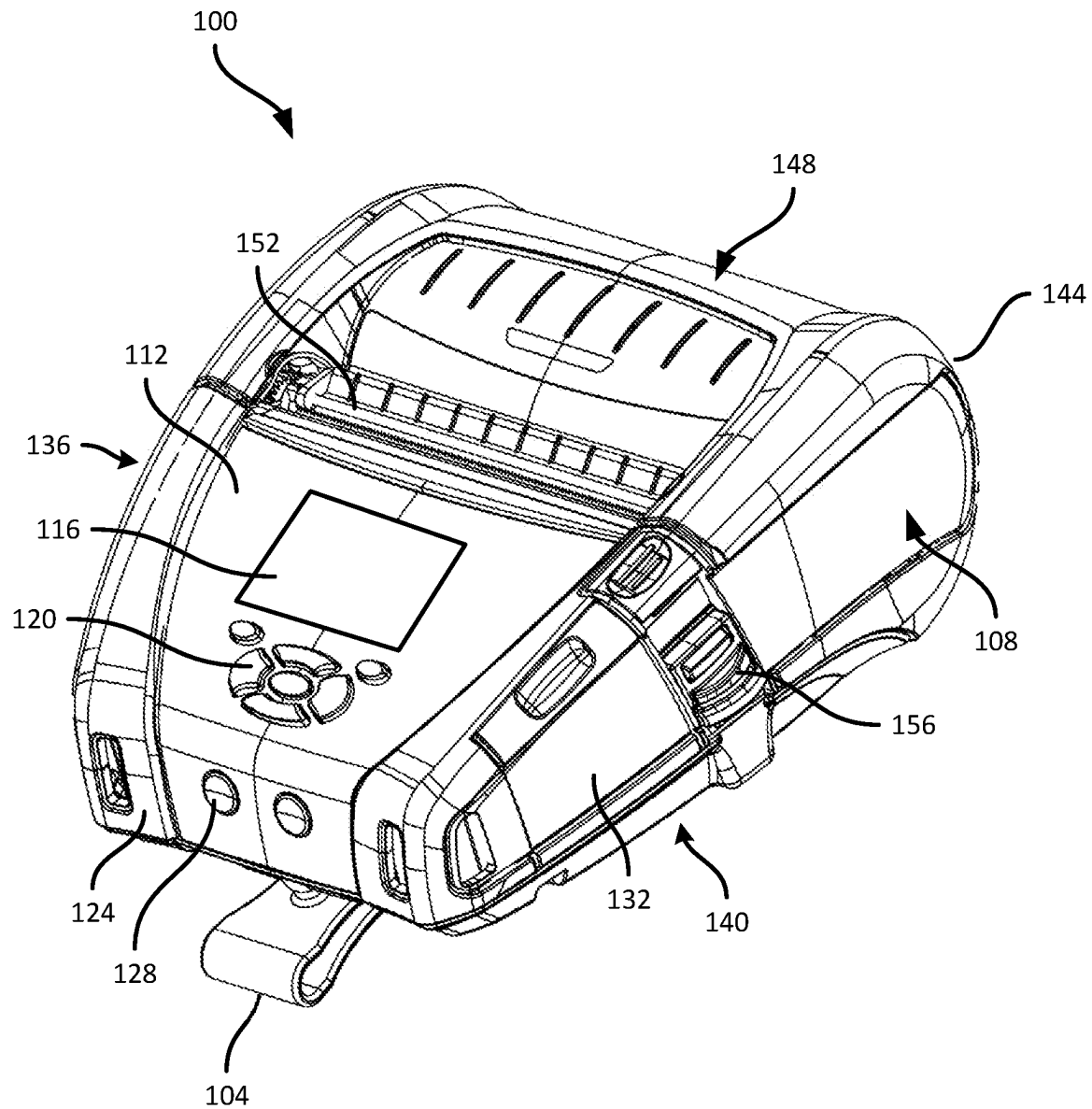


FIG. 1

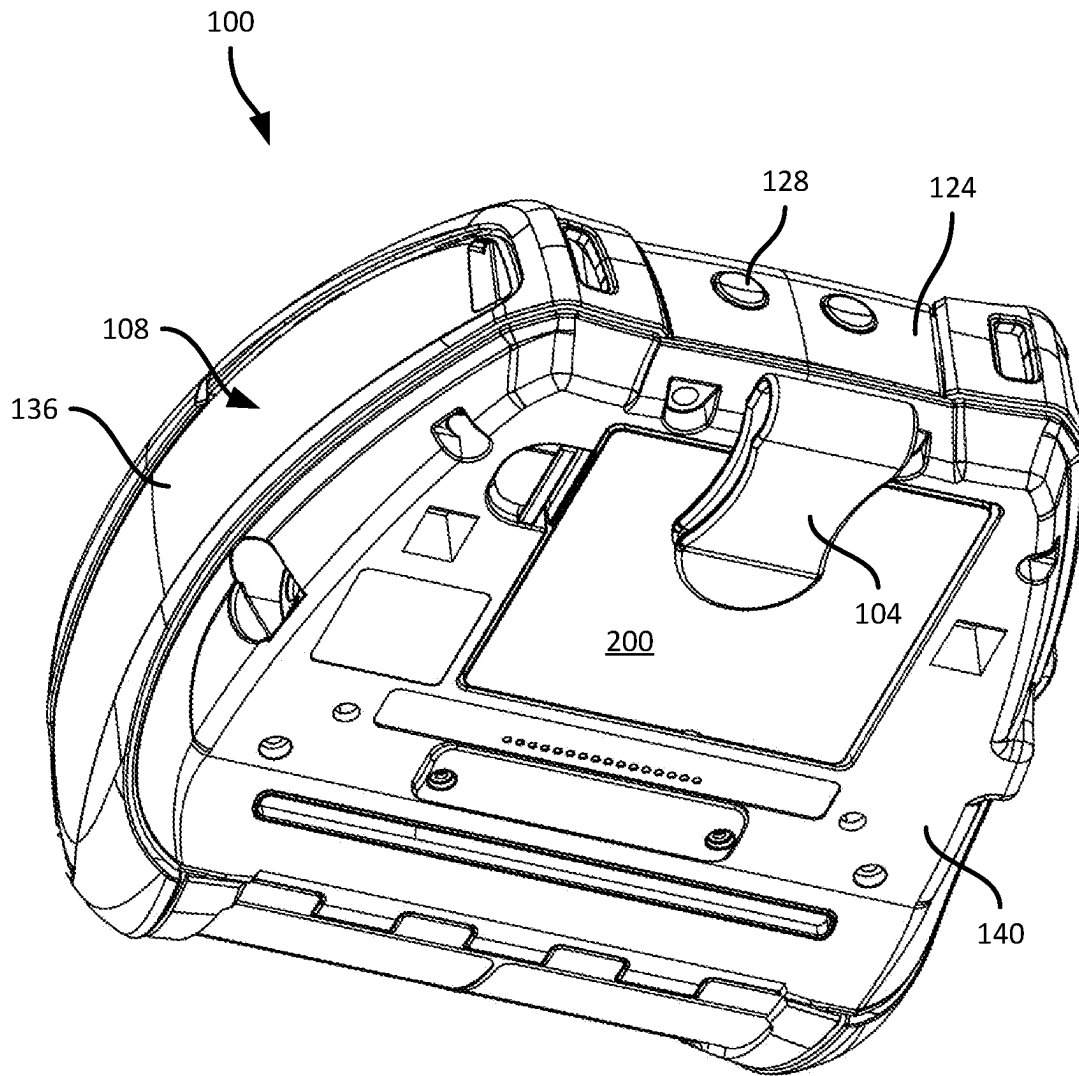


FIG. 2

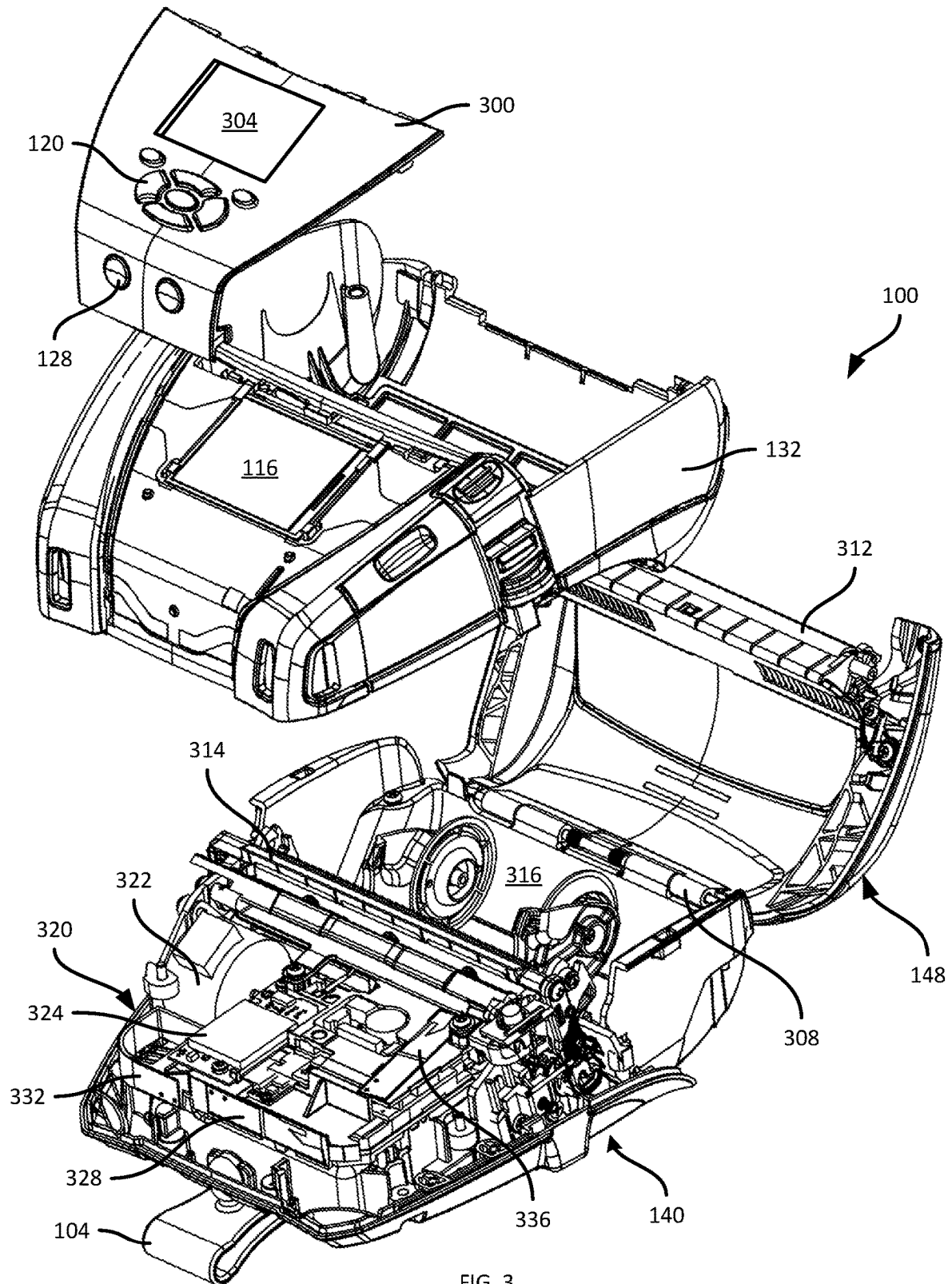


FIG. 3

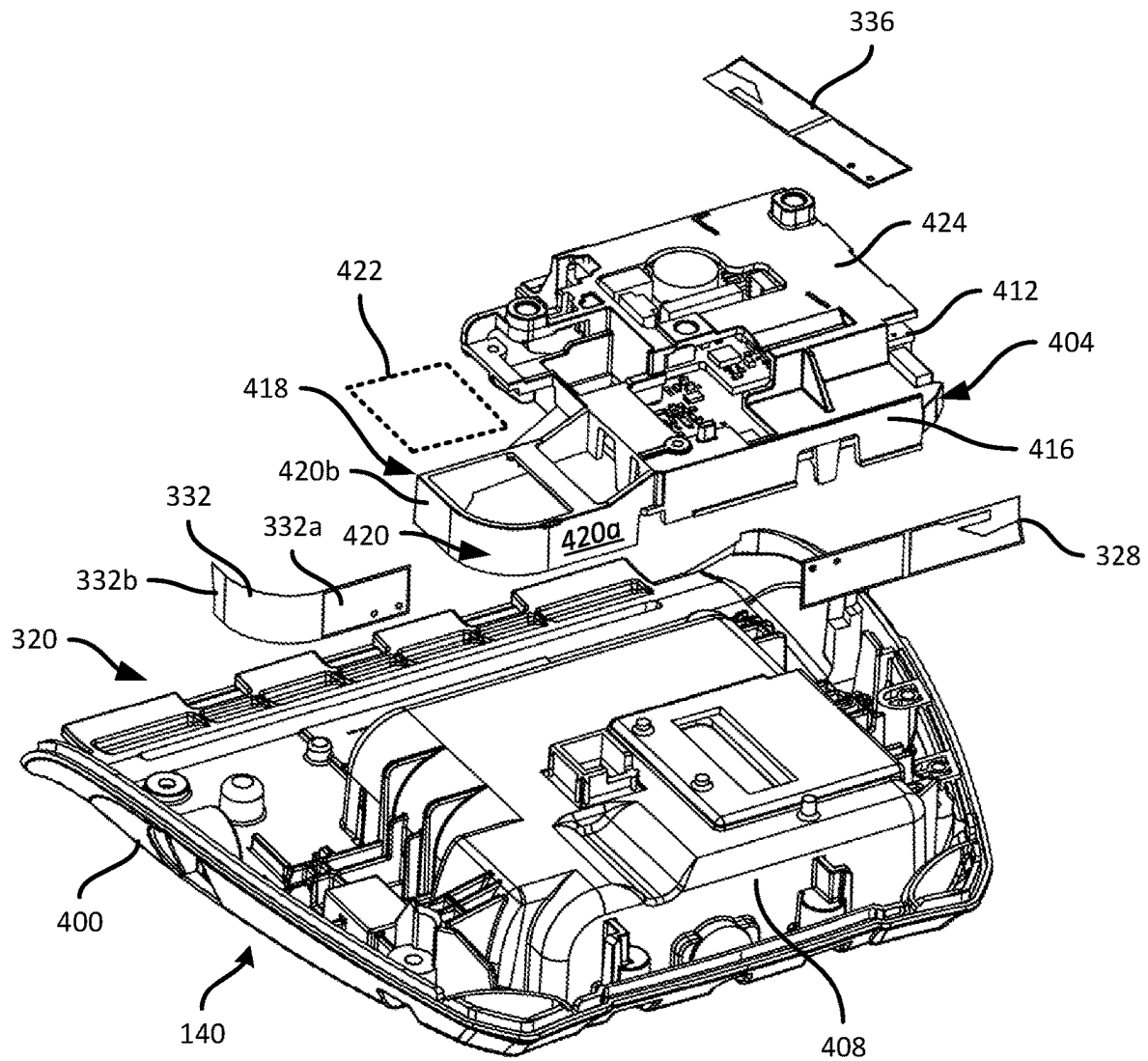


FIG. 4

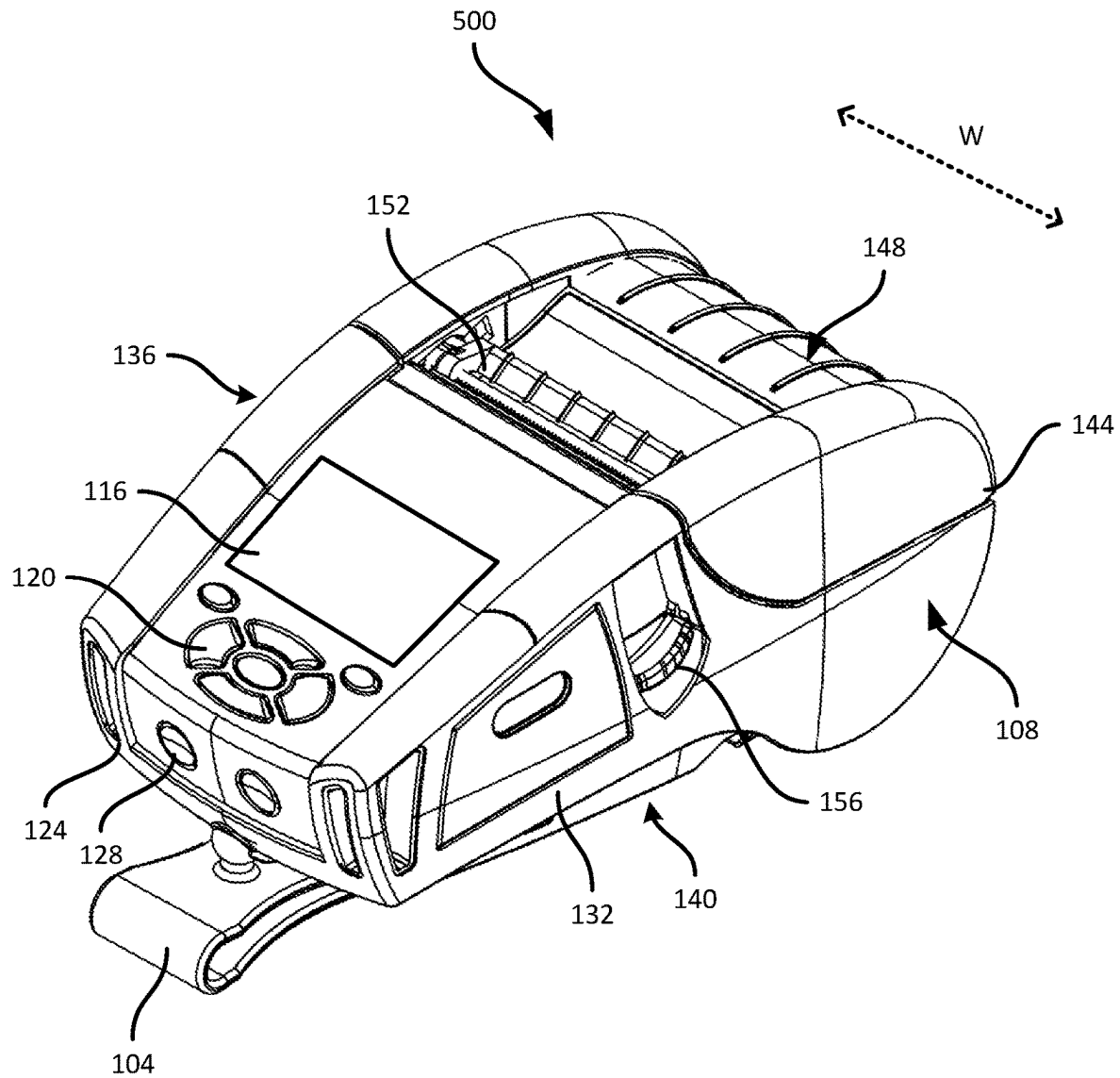
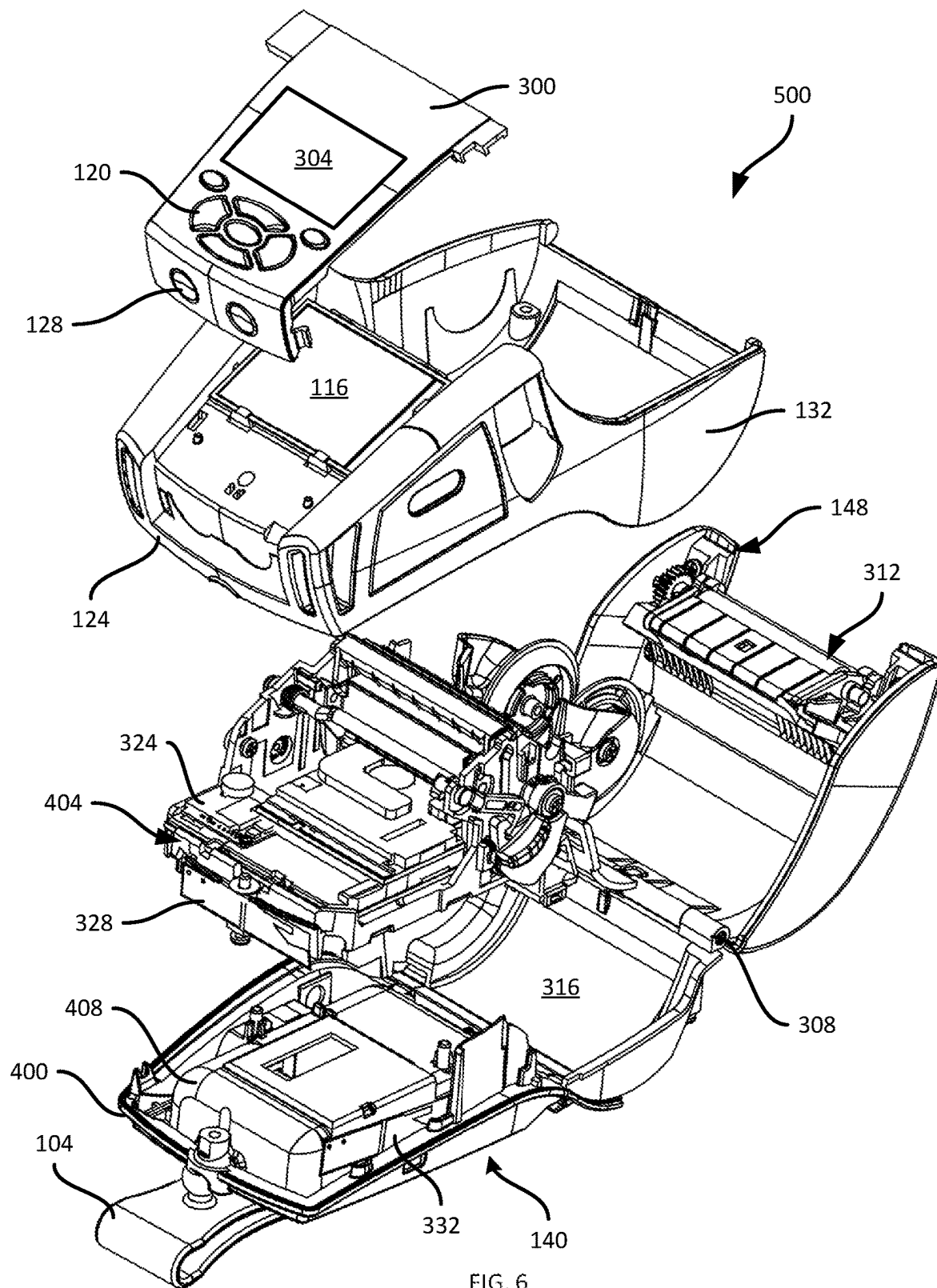


FIG. 5



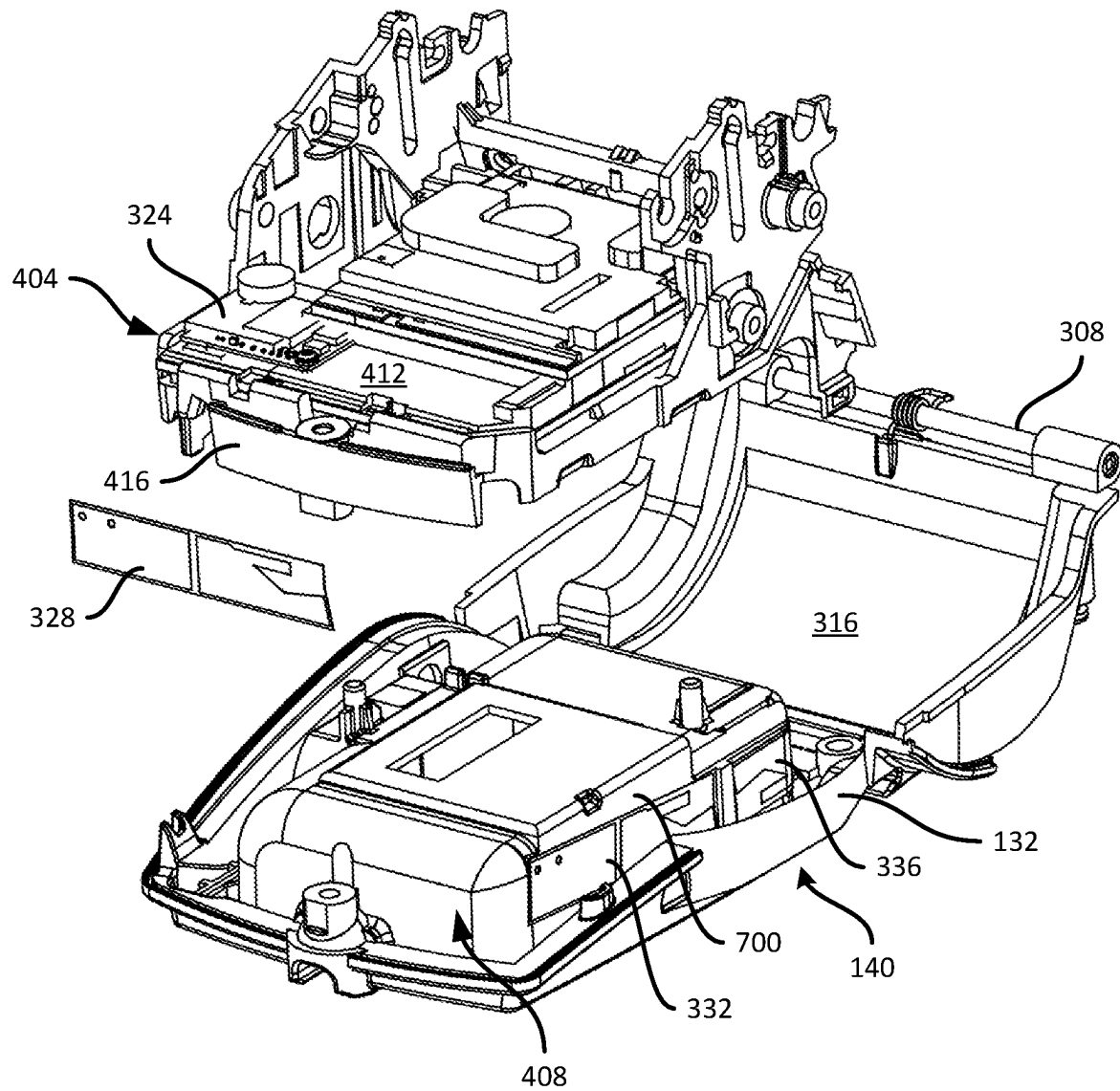


FIG. 7

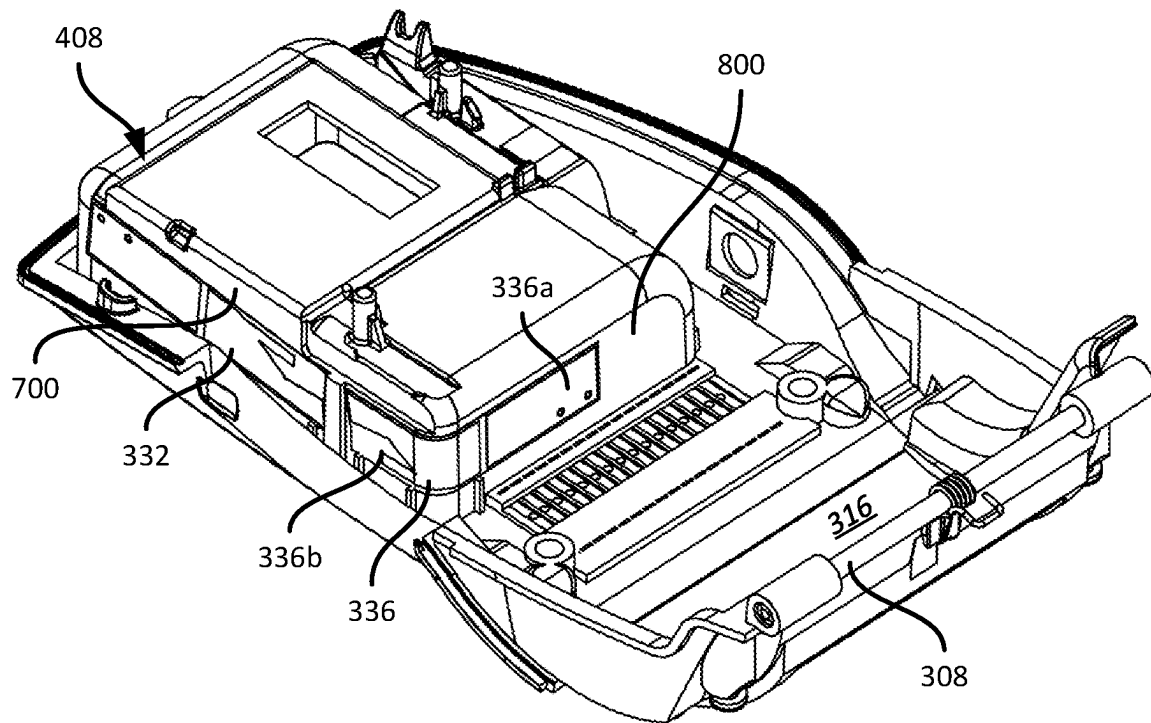


FIG. 8

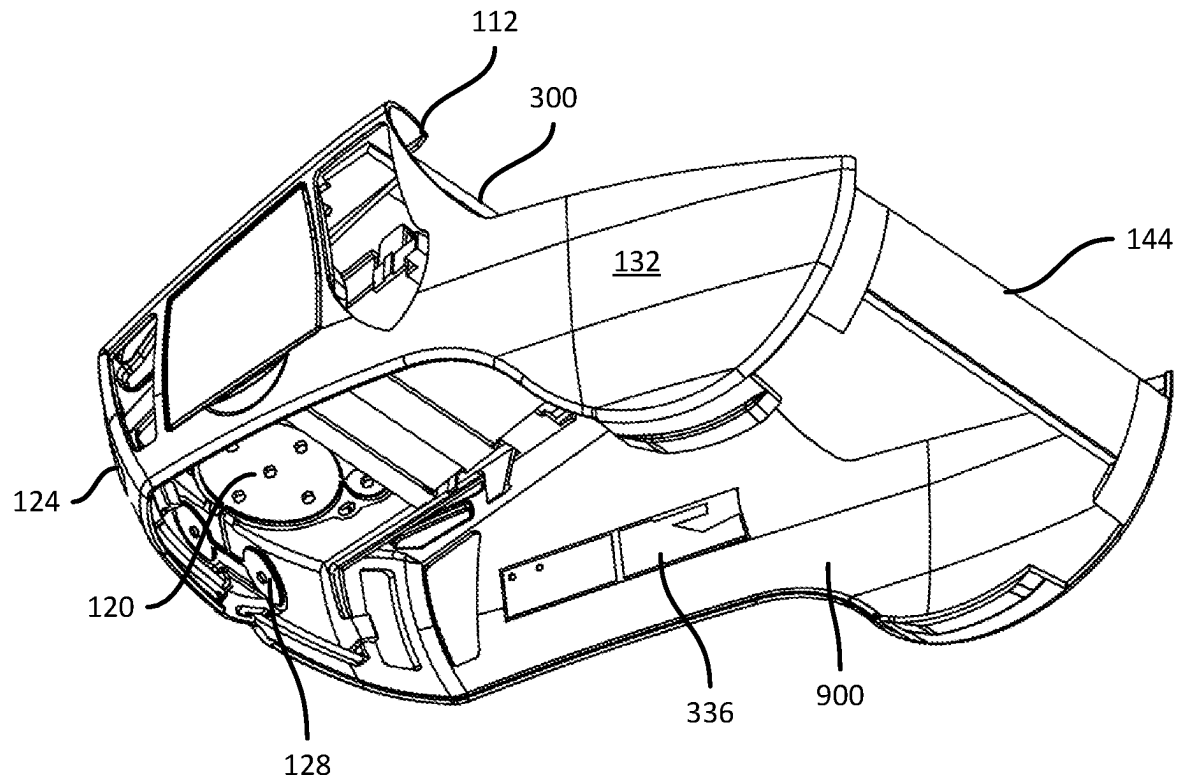


FIG. 9

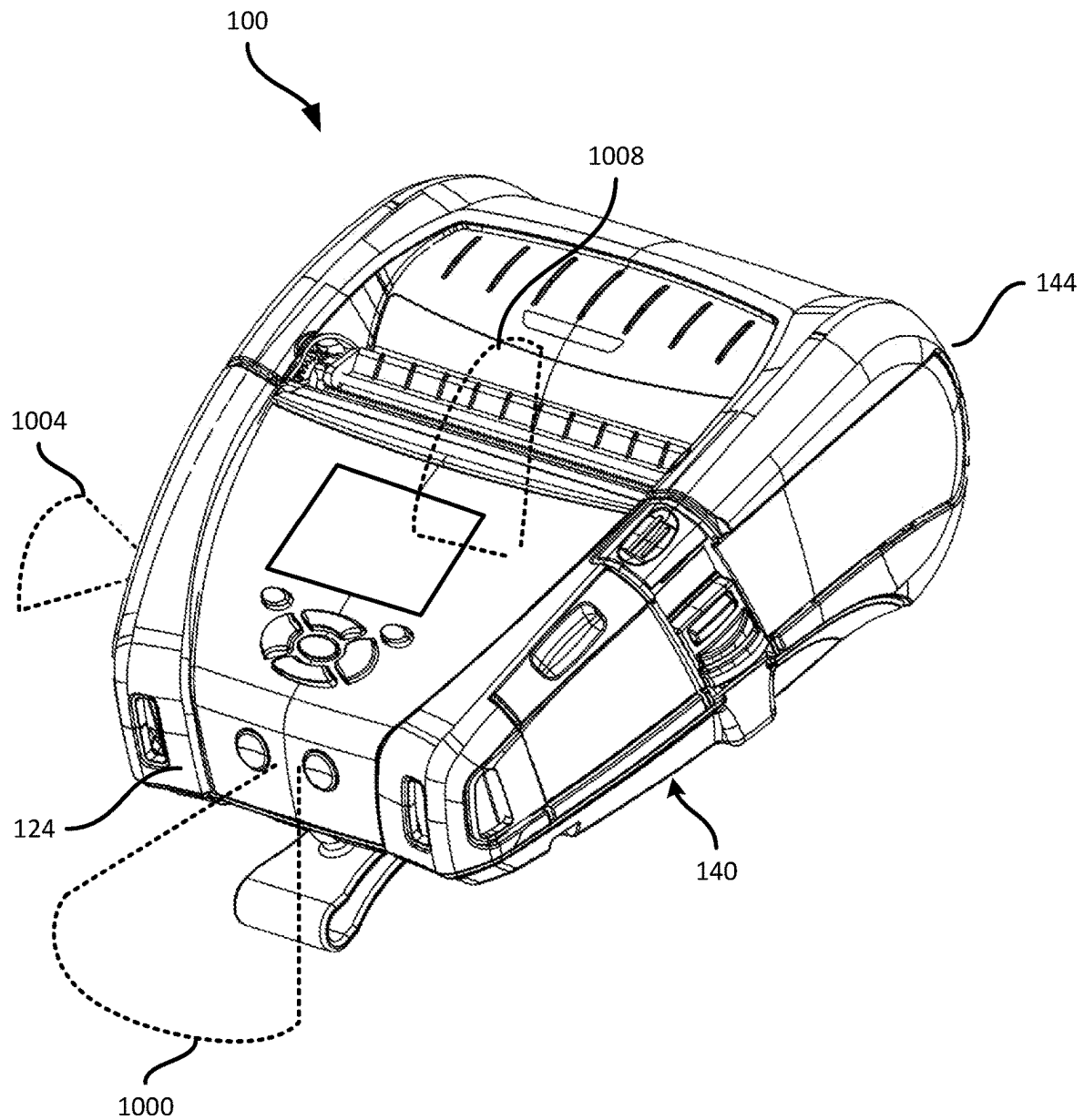


FIG. 10

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MULTI-ANTENNA MOBILE PRINTERS**BACKGROUND**

Media processing devices, such as mobile printers, may communicate using more than one communications standard and/or frequency. Providing components within a media processing device to implement multiple communications standards and/or frequencies may be complicated by limited physical space in which to place the various components of the device and by potential radiofrequency interference caused by the various components.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

FIG. 1 is a top perspective view of a media processing device.

FIG. 2 is bottom perspective view of the media processing device of FIG. 1.

FIG. 3 is an exploded view of the media processing device of FIG. 1.

FIG. 4 is a diagram of a portion of the interior of the media processing device of FIG. 1.

FIG. 5 is another perspective view of a media processing device.

FIG. 6 is an exploded view of the media processing device of FIG. 5.

FIG. 7 is an exploded view of a portion of the interior of the media processing device of FIG. 5.

FIG. 8 is a perspective view of a back portion of a further media processing device in isolation.

FIG. 9 is a diagram of an interior of a portion of a housing of the further media processing device.

FIG. 10 is a perspective view of the media processing device of FIG. 1, illustrating radiation patterns for the antennas of the media processing device.

Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

Examples disclosed herein are directed to a media processing device, comprising: a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment; a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open

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position to expose the media supply chamber; a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing for dispensing the media; a frame supported by the housing within the compartment, the frame defining (i) a first mounting surface adjacent to a second end of the housing opposite the first end, the first mounting surface oriented towards the second end, (ii) a second mounting surface, a portion of the second mounting surface orthogonal to the first mounting surface and oriented towards a side of the housing, and (iii) a third mounting surface oriented towards the front wall of the housing; a first antenna disposed on the first mounting surface; a second antenna disposed on the second mounting surface, controllable with the first antenna to implement a first communications standard; and a third antenna disposed on the third mounting surface, and controllable to implement a second communications standard.

Additional examples disclosed herein are directed to a media processing device, comprising: a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment; a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open position to expose the media supply chamber; a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing opposite the back wall, for dispensing the media; a frame supported by the housing within the compartment, the frame defining a first mounting surface adjacent to a second end of the housing opposite the first end, the first mounting surface oriented towards the second end; a plurality of secondary walls defining a battery well extending into the compartment from a back wall of the housing, the secondary walls defining (i) a second mounting surface within the compartment, orthogonal to the first mounting surface and oriented towards a side of the housing, and (ii) a third mounting surface within the compartment having a portion oriented towards the side of the housing; a first antenna disposed on the first mounting surface; a second antenna disposed on the second mounting surface, controllable with the first antenna to implement a first communications standard; and a third antenna disposed on the third mounting surface, and controllable to implement a second communications standard.

Further examples disclosed herein are directed to a media processing device, comprising: a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment; a first side wall of the housing having an inner surface within the compartment defining a first mounting surface; a first antenna disposed on the first mounting surface of the side wall, the first antenna controllable to implement a first communications standard; a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open position to expose the media supply chamber; a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing opposite the back wall, for dispensing the media; a frame supported by the housing within the compartment, the frame defining a second mounting surface adjacent to a second end of the housing opposite the first end, the second mounting surface oriented towards the second end; a second antenna disposed on the second

mounting surface; an internal wall disposed within the compartment, the internal wall defining a third mounting surface adjacent to a second side wall of the housing, opposite the first side wall, and facing towards the second side wall; and a third antenna disposed on the third mounting surface, controllable with the second antenna to implement a second communications standard.

FIG. 1 illustrates a media processing device **100**, such as a label printer (also referred to herein simply as the printer **100**). The printer **100** can be implemented as a battery-powered mobile label printer, as illustrated. For example, the printer **100** can be worn or otherwise carried by an operator via a mounting accessory **104**, such as a belt clip (as illustrated in FIG. 1), a carrying strap, or the like, coupled to a housing **108** of the printer **100**. The printer **100** can also be implemented in a wide variety of other form factors, including a desktop printer or the like.

The printer **100** includes various components configured to apply indicia to media such as discrete labels, a continuous paper strip, identity cards, or the like. In the illustrated example, the printer **100** applies indicia via direct thermal printing, in which a print head applies heat to portions of the media, such as thermally-sensitive paper or other media, to activate a thermochromic coating of the media and generate indicia thereon. In other examples, the printer **100** can implement thermal transfer printing, or the like. In some examples, the media processing device **100** can include a radio frequency identification (RFID) assembly configured to write data to RFID tags embedded in labels or other media, in addition to or instead of applying indicia to the media.

The printer **100** includes the above-mentioned housing **108**, which includes a plurality of walls. The walls of the housing **108** include, in this example, a front wall **112**, e.g., configured to face outwards from an operator when the printer **100** is worn or otherwise carried by the operator. The front wall **112** can include an opening in which a display **116** is disposed, and can also support control elements such as buttons **120**, e.g., for navigating menus or other information presented on the display **116**.

The walls of the housing **108** can also include an end wall **124**, defining one end of the printer **100** (e.g., the end facing upwards when the printer **100** is worn or otherwise carried by an operator). The end wall **124** can support additional control elements, such as buttons **128** (e.g., a power button and a feed button, although a wide variety of other controls can be implemented by the buttons **128**). The walls of the housing **108** further include side walls **132** and **136**, extending between the front wall **112** and an opposing back wall, as well as between the end wall **124** and an opposing end **144** of the printer **100**.

The printer **100** further includes a cover **148** movably coupled to the housing **108** at the end **144**. The cover **148**, as discussed below, is movable between a closed position, illustrated in FIG. 1, and an open position that exposes certain internal portions of the printer **100**. The cover **148** also supports a platen roller that cooperates with a print head supported by the housing **108**, adjacent to a portion of the front wall **112**, to form an outlet **152** for dispensing media processed via the print head. The cover **148** can be unlatched or otherwise released to move relative to the housing **108** via one or more actuators such as a latch **156**.

Turning to FIG. 2, the printer **100** is shown from the opposite side as in FIG. 1, revealing the other side wall **136** and the back wall **140**. The housing **108** also defines a battery well extending into the back wall **140**, in which a battery pack **200** (e.g., a rechargeable battery) can be remov-

ably inserted. In the present example, removal or insertion of the battery pack **200** may also involve moving or removing the mounting accessory **104** from the housing **108**.

The printer **100** can transmit and receive data, such as data defining indicia to be applied to media, as well as a wide variety of other information (e.g., firmware updates, usage statistics, and the like), by wireless communications with other computing devices. To that end, the printer **100** may support two or more wireless communications standards. For example, the printer **100** can support a personal area network (PAN) standard, such as Bluetooth™ (e.g., Bluetooth 5.2, although more recent and/or older versions may also be supported), Bluetooth Low Energy, and the like. The printer **100** can also support a wireless local area network (WLAN) standard, such as one or more members of the Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards (e.g., 802.11 ax, also referred to as Wi-Fi 6 or Wi-Fi 6E; various other 802.11 WLAN standards may also be supported). The printer **100** therefore includes various hardware components for supporting the above communications standards. In some examples, the printer **100** includes a first antenna for PAN communications, and two additional antennas for WLAN communications, e.g., to support 2x2 multiple-input and multiple-output (MIMO) WLAN communications.

The above-mentioned antennas may be susceptible to interference from other internal components of the printer **100**, as well as from each other. Further, optimal performance of the antennas may involve, for example, placing the two WLAN antennas at substantially orthogonal orientations to one another. In order to accommodate the antennas in the limited physical space available within the printer **100**, while mitigating either or both of performance impacts on the antennas by other components within the printer **100**, and increased assembly complexity, the printer **100** defines various mounting surfaces for the antennas, as described below.

Turning to FIG. 3, a partially exploded view of the printer **100** is shown, in which a portion of the housing **108** defining the side walls **132** and **136**, as well as the front wall **112**, is separated from another portion of the housing **108** defining the back wall **140**. A portion **300** of the front wall **112** is shown separate from the remainder of the front wall **112**, illustrating an opening **304** for the display **116**.

The cover **148** is illustrated in the open position in FIG. 3, e.g., having rotated relative to the housing **108** about an axis **308**. As mentioned earlier, the cover **148** includes a platen roller **312**, and the housing **108** supports a print head **314**, e.g., a thermal print head. The platen roller **312** is configured to draw media from a media supply chamber **316** defined by the housing **108**. The media supply chamber **316** can support, for example, a roll of paper or other media that is fed to the print head **314** for application of indicia and dispensing via the outlet **152**. The cover **148** exposes the media supply chamber **316** in the open position as shown in FIG. 3, e.g., for installation of a media supply, and encloses the media supply chamber **316** in the closed position, as shown in FIG. 1.

The housing **108** also defines a compartment **320** adjacent to the media supply chamber **316**, and enclosed within the housing **108**. The compartment **320** contains a variety of other components of the printer **100**, including for example a controller **324** and associated circuits, chips, and the like, for controlling the operation of the printer **100**, communicating with other computing devices, and the like. The compartment **320** further contains the print head **314**, and can also include a drive assembly including a motor **322**

configured to drive the platen roller **312**, e.g., via a gear train supported at least partly within the compartment **320** (portions of the gear train may extend into the media supply chamber **316**).

The compartment **320** also contains the above-mentioned antennas, including a first antenna **328**, a second antenna **332**, and a third antenna **336** such that the antennas **328**, **332**, and **336** are included within the housing **108** and radiofrequency signals radiate through the housing **108** to and from the antennas **328**, **332**, and **336**. In this example, the first and second antennas **328** and **332** can implement a first communications standard, such as a WLAN standard. The first and second antennas **328** and **332** can be controlled together, for example, for 2x2 MIMO communications. The third antenna **336** can implement a second communications standard, such as a PAN standard. As discussed below, the antennas **328**, **332**, and **336** are disposed on mounting surfaces defined within the compartment **320** to place radiative portions of the antennas **328** and **332** in substantially orthogonal orientations relative to one another, as well as to mitigate interference between antennas, and mechanical and/or radiofrequency interference with antenna performance by other components of the printer **100**. The antennas **328**, **332**, and **336** can be implemented as patch antennas, although the antennas **328**, **332**, and **336** can be implemented using different types of antennas.

Turning to FIG. 4, a portion of the housing **108** is shown in isolation. In particular, a wall **400** of the housing **108**, defining part of the back wall **140**, is illustrated. As shown in FIG. 4, the printer **100** includes a frame **404** disposed within the compartment **320**, e.g., resting on or otherwise coupled to the wall **400**. In this example, the housing **108** includes a set of secondary walls defining a battery well **408**, as noted earlier. The secondary walls extend into the compartment **320** from the wall **400** (e.g., from the back wall **140**), defining a cavity outside the compartment **320** into which the battery pack **200** can be inserted. In this example, the frame **404** is supported within the compartment **320** on at least one of the walls of the battery well **408**.

The frame **404** supports various components of the printer **100**, and some or all of those components, including the antennas **328**, **332**, and **336**, can be assembled with the frame **404** prior to installation of the frame **404** into or onto the housing **108** (e.g., by fastening the frame **404** to the battery well **408**). Assembly of the printer **100** may therefore be simplified, e.g., in comparison to an embodiment in which one or more of the antennas **328**, **332**, and **336** are installed on portions of the housing **108**, or on movable parts such as the cover **148**.

The components supported by the frame **404** can include one or more printed circuit boards **412** carrying the controller **324** (not shown in FIG. 4) and other electronic elements of the printer **100**. The frame **404** also defines, in this example, mounting surfaces for the antennas **328**, **332**, and **336**. In particular, the frame **404** defines a first mounting surface **416** upon which the antenna **328** is disposed. The first mounting surface **416** is adjacent to the end wall **124** (e.g., the opposite end from the end at which the cover **148** is mounted to the housing **108**), and is oriented towards the end wall **124**. That is, the mounting surface **416** faces outwards from the compartment **320** towards the exterior of the printer **100**. The antenna **328**, in turn, is configured to radiate in the direction in which the mounting surface **416** faces. The antenna **328** is therefore placed closer to the end wall **124** than to the media supply chamber **316**, the platen roller **312**, and the print head **314**. The antenna **328** can be separated from the media supply chamber **316**, the platen

roller **312**, and the print head **314** by the frame **404** and the other components thereon, in other words. Further, the antenna **328** radiates through the end wall **124**, rather than through the circuit board **412** and other components between the mounting surface **416** and the end **144** of the printer **100**. That is, the antenna **328** radiates substantially away from the platen roller **312**, the print head **314**, and the media supply chamber **316**. As seen from FIGS. 3 and 4, the first mounting surface **416** can face towards the buttons **128**. Although the antenna **328** therefore radiates through the buttons **128**, the buttons **128** contain little conductive material and therefore are not expected to interfere significantly with the antenna **328**.

The frame **404** also includes a projection **418** that defines a second mounting surface **420**, on which the second antenna **332** is disposed. The projection **418** places the mounting surface **420** adjacent to the side wall **136**, and defines, with a remainder of the frame, a space **422** that accommodates the motor **322** seen in FIG. 3. The mounting surface **420** wraps around a corner of the frame **404**, and thus includes a portion **420a** that is oriented towards the end wall **124** (and is substantially coplanar with the mounting surface **416**), and another portion **420b** that is oriented towards the side wall **136**, substantially orthogonal to the mounting surface **416**. The antenna **332**, in turn, includes a feed segment or portion **332a**, configured to be disposed on the portion **420a** of the mounting surface **420**, and a radiative portion **420b** configured to be disposed on the portion **420b** of the mounting surface **420**. More generally, each of the antennas **328**, **332**, and **336** includes a feed portion and a radiative portion. The feed segment **332a** includes ports (e.g., a transmission port and a reception port) for connection to the controller **324** or another suitable component, while the radiative segment **332b** is configured to radiate, e.g., towards the side wall **136** rather than towards the side wall **132** (and through other components such as the controller **324**, circuit board **412**, and the like). The antenna **332**, in other words, is positioned to radiate in a direction substantially orthogonal to the radiative direction of the antenna **328**. The antenna **332** is further positioned so as to be separated from the media supply chamber **316**, the platen roller **312**, and the print head **314** by either or both of the motor **322** and the frame **404**. As seen in FIG. 3, the antenna **332** is also separated from the media supply chamber **316** and the platen roller **312** by the print head **314**, and radiates substantially away from each of the platen roller **312**, the print head **314**, and the media supply chamber **316**.

The frame **404** further defines a third mounting surface **424**, on which the antenna **336** is disposed. The mounting surface **424** is oriented towards the front wall **112** of the housing **108**, and therefore away from the battery well **408**. As seen in FIG. 4, the mounting surface **424** is on an upper portion of the frame **404**, placing the circuit board **412** in between the mounting surface **424** and the battery well **408**. The antenna **336** is configured to radiate towards the front wall **112**, and away from the circuit board **412** and battery well. Placement of the antenna **336** on the mounting surface **424** thus mitigates interference with performance of the antenna **336** by the circuit board **412**, the battery pack **200**, and the like. Placement of the antenna **336** on the mounting surface **424** also distances the antenna **336** sufficiently from the antennas **328** and **332** to mitigate interference between the antenna **336** and the other antennas. Further, as seen from FIGS. 3 and 4, the mounting surface **424** is adjacent to the side wall **132**, and specifically is closer to the side **132** than the display panel **116**. The antenna **336** can therefore radiate through the front wall **112** and not through the display **116**,

avoiding or at least reducing interference that could result from radiating through the display 116. As seen in FIG. 3 and FIG. 4, the antenna 336 is closer to the media supply chamber 316, and to the print head 314 (and the platen roller 312, when the cover 148 is closed) than the antennas 328 and 332. However, the antenna 336 remains separated from the platen roller 312 and the media supply chamber 316 by the print head 314, and radiates substantially away from each of the platen roller 312, the print head 314, and the media supply chamber 316.

Turning to FIG. 5, a printer 500 is shown according to another embodiment. In the discussion below, like numbers to those discussed above refer to components that are not necessarily identical, but are functionally similar to those discussed above. Certain features of those components, such as physical dimensions, shapes, and positions, may vary in the printer 500 relative to the printer 100, as shown in the drawings and discussed below.

As seen in FIG. 5, a width W of the printer 500 (e.g., the distance between the side walls 132 and 136) may be reduced in comparison to the printer 100 shown in FIGS. 1-4. The interior of the printer 500 may therefore have a smaller volume. As discussed below, the printer 500 provides mounting surfaces for the antennas 328, 332, and 336 that are positioned differently than those of the printer 100.

Turning to FIG. 6, an exploded view of the printer 500, in which the frame 404 is shown removed from the wall 400 (which, as noted in connection with FIG. 4, defines a portion of the back wall 140), and the side walls 132 and 136, as well as the front wall 112 are removed from the remainder of the housing 108. As seen in FIG. 6, the reduced width of the printer 500 compared to the printer 100 results in the display 116 occupying a larger fraction of the width of the printer 500. As a result, placing the antenna 336 on an upper surface of the frame 404 as shown in FIG. 4 may result in the antenna 336 being located underneath the display 116, rather than underneath a portion of the front wall 112 beside the display 116. As discussed below, the antenna 336, which is not visible in FIG. 6, is installed in a location distinct from the frame 404. Further, for example as a result of the reduced size of the frame 404 and/or the reduced volume for accommodating other components of the printer (e.g., gear trains, motors, and the like), the antenna 332 is also installed on a mounting surface distinct from the frame 404. The antenna 328 remains mounted on the frame 404.

Referring to FIG. 7, the frame 404 and the back wall 140 are shown in isolation from the remainder of the housing 108. As seen in FIG. 7, the first mounting surface 416, as described earlier in connection with FIG. 4, is oriented towards the end wall 124, away from the media supply chamber 316. The antenna 328 can therefore radiate through the end wall 124 and the buttons 128. As also seen in FIG. 7, the antenna 332 and the antenna 336 are disposed on internal walls, located within the compartment 320. In this example, mounting surfaces for the antennas 332 and 336 are defined by the battery well 408. In particular, the antenna 332 and the antenna 336 are disposed on a side wall 700 of the battery well 408, oriented towards the side wall 132, such that the antennas 332 and 336 radiate away from the battery well, through the side wall 132.

As shown in FIG. 8, while the second mounting surface for the antenna 332 is located entirely on the side wall 700 of the secondary walls defining the battery well 408, the third mounting surface for the antenna 336 wraps from the side wall 700 to an end wall 800, facing the media supply chamber 316. The antenna 336 includes a feed segment 336a (that is, a non-radiating segment) on the end wall 800, and

a radiating segment 336b on the side wall 700. As such, the radiating segment 336b can be oriented such that the radiating segment 336b faces the media supply chamber 316 and the radiating segment may receive and/or transmit radiofrequency waves through the media supply chamber 316 as well as any media contained within the media supply chamber 316.

The positions of mounting surfaces and the antennas 328, 332, and 336 as described above can also be combined in other examples. For instance, in a further example printer, the antenna 336 can be disposed on the battery well 408 as in the printer 500, while the antennas 328 and 332 are disposed as shown in connection with the printer 100. Further variations are also contemplated. Another example variation is shown in FIG. 9, in which a portion of the housing 108 is shown in isolation. In particular, the side walls 132 and 136, as well as the upper wall 112 (including the portion 300) and the end wall 124, are illustrated in isolation from the remainder of the housing 108. That is, the back wall 140, as well as the frame 404 and internal components supported on the frame 404, are omitted from FIG. 9.

While the antennas 328 and 332 can be installed as shown in FIGS. 7 and 8 in this example (or, in other examples, in a combination of the orientations shown in FIGS. 1 to 8), the antenna 336 in the example of FIG. 9 is disposed on a mounting surface that is not on the battery well 408 or the frame 404. Instead, in this example the mounting surface is defined on the side wall 136, e.g., on an inner surface 900 of the side wall 136. The mounting surface for the antenna 336 is therefore oriented into the compartment 320, although the antenna 336 can be mounted to radiate outwards, through the side wall 136. The mounting surface for the antenna 336 may therefore, in this example, be substantially parallel to the mounting surface for the antenna 332 (e.g., on the wall 700 of the battery well 408), but the mounting surfaces for the antennas 332 and 336 can be located adjacent to opposite sides of the housing 108, and the antennas 332 and 336 can radiate in substantially opposite directions.

As will be apparent in the discussion above, the mounting surfaces implemented to support the antennas 328, 332, and 336 position the antennas to reduce or avoid radiating through components such as the display 116, the print head 314, the circuit board 412, the battery, and the like (e.g., large conductive elements that can interfere with antenna performance). Further, as shown in FIG. 10, for example, the antennas 328, 332, and 336 may avoid radiating through the back wall 140 or the end 144, as the back wall 140 may face an operator when the printer 100 is in use, and the end 144 may face the ground, floor, or the like when the printer is in use. As shown in FIG. 10, the antenna 328 can radiate in a region 1000, e.g., directed outwards from the end wall 124. The antenna 332 can radiate in a region 1004, e.g., substantially orthogonal to the region 1000, and the antenna 336 can radiate in a region 1008, which can be substantially orthogonal to the regions 1000 and 1004.

Further, installation of the antennas 328, 332, and 336 as discussed above may simplify assembly of the printers 100, 500, and the like, as the 328, 332, and 336 are not disposed on moving parts, and with the exception of the embodiment shown in FIG. 9, can be installed and connected to the controller 324 independently of the side walls 132, 136, and the front wall 112.

In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as

set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.

The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

Certain expressions may be employed herein to list combinations of elements. Examples of such expressions include: “at least one of A, B, and C”; “one or more of A, B, and C”; “at least one of A, B, or C”; “one or more of A, B, or C”. Unless expressly indicated otherwise, the above expressions encompass any combination of A and/or B and/or C.

It will be appreciated that some embodiments may be comprised of one or more specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used.

Moreover, an embodiment can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g.,

comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

The invention claimed is:

1. A media processing device, comprising:

- a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment;
- a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open position to expose the media supply chamber;
- a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing for dispensing the media;
- a frame supported by the housing within the compartment, the frame defining (i) a first mounting surface adjacent to a second end of the housing opposite the first end, the first mounting surface oriented towards the second end, (ii) a second mounting surface, a portion of the second mounting surface orthogonal to the first mounting surface and oriented towards a side of the housing, and (iii) a third mounting surface oriented towards the front wall of the housing;
- a first antenna disposed on the first mounting surface;
- a second antenna disposed on the second mounting surface, controllable with the first antenna to implement a first communications standard; and
- a third antenna disposed on the third mounting surface, and controllable to implement a second communications standard.

2. The media processing device of claim 1, wherein the front wall defines an opening therethrough; wherein the media processing device further comprises a display panel supported in the opening; and wherein the third mounting surface faces a portion of the front wall distinct from the opening.

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3. The media processing device of claim 1, further comprising a printed circuit board supported by the frame, the printed circuit board carrying at least one communications controller for controlling the first, second, and third antennas;

wherein the third mounting surface is between the printed circuit board and the front wall.

4. The media processing device of claim 1, further comprising:

a control input supported by the housing at the first end; wherein the first mounting surface is oriented towards the control input.

5. The media processing device of claim 1, wherein a further portion of the second mounting surface is substantially coplanar with the first mounting surface; and

wherein a radiative segment of the second antenna is disposed on the portion of the second mounting surface, and a feed segment of the second antenna is disposed on the further portion of the second mounting surface.

6. The media processing device of claim 1, wherein the housing defines a battery well extending into the compartment from a back wall of the housing opposite the front wall; and

wherein the third mounting surface is oriented away from the battery well.

7. A media processing device, comprising:

a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment;

a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open position to expose the media supply chamber;

a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing opposite the back wall, for dispensing the media;

a frame supported by the housing within the compartment, the frame defining a first mounting surface adjacent to a second end of the housing opposite the first end, the first mounting surface oriented towards the second end;

a plurality of secondary walls defining a battery well extending into the compartment from a back wall of the housing, the secondary walls defining (i) a second mounting surface within the compartment, orthogonal to the first mounting surface and oriented towards a side of the housing, and (ii) a third mounting surface within the compartment having a portion oriented towards the side of the housing;

a first antenna disposed on the first mounting surface;

a second antenna disposed on the second mounting surface, controllable with the first antenna to implement a first communications standard; and

a third antenna disposed on the third mounting surface, and controllable to implement a second communications standard.

8. The media processing device of claim 7, wherein the portion of the third mounting surface and the second mounting surface are defined on a side wall of the secondary walls, the side wall facing the side of the housing;

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wherein a further portion of the third mounting surface wraps onto an end wall of the secondary walls; and wherein a radiative segment of the third antenna is disposed on the portion of the third mounting surface, and a feed segment of the third antenna is disposed on the further portion of the third mounting surface.

9. The media processing device of claim 8, wherein the end wall of the secondary walls is oriented towards the media supply chamber.

10. The media processing device of claim 7, further comprising:

a control input supported by the housing at the first end; wherein the first mounting surface is oriented towards the control input.

11. A media processing device, comprising:

a housing having a plurality of walls enclosing a compartment, and defining a media supply chamber adjacent to the compartment;

a first side wall of the housing having an inner surface within the compartment defining a first mounting surface;

a first antenna disposed on the first mounting surface of the side wall, the first antenna controllable to implement a first communications standard;

a cover movably coupled to a first end of the housing adjacent to the media supply chamber, the cover movable between a closed position to enclose the media supply chamber, and an open position to expose the media supply chamber;

a print head configured to (i) apply indicia to media from the media supply chamber, and (ii) cooperate with the cover to form an outlet at a front wall of the housing opposite the back wall, for dispensing the media;

a frame supported by the housing within the compartment, the frame defining a second mounting surface adjacent to a second end of the housing opposite the first end, the second mounting surface oriented towards the second end;

a second antenna disposed on the second mounting surface;

an internal wall disposed within the compartment, the internal wall defining a third mounting surface adjacent to a second side wall of the housing, opposite the first side wall, and facing towards the second side wall; and

a third antenna disposed on the third mounting surface, controllable with the second antenna to implement a second communications standard.

12. The media processing device of claim 11, wherein the first communications standard is a wireless personal area network standard; and

wherein the second communications standard is a wireless local area network standard.

13. The media processing device of claim 11, further comprising:

a control input supported by the housing at the first end; wherein the first mounting surface is oriented towards the control input.

14. The media processing device of claim 11, wherein the first mounting surface is substantially parallel to the second mounting surface.

15. The media processing device of claim 11, further comprising a battery well extending into the compartment; wherein the internal wall forms a portion of the battery well.